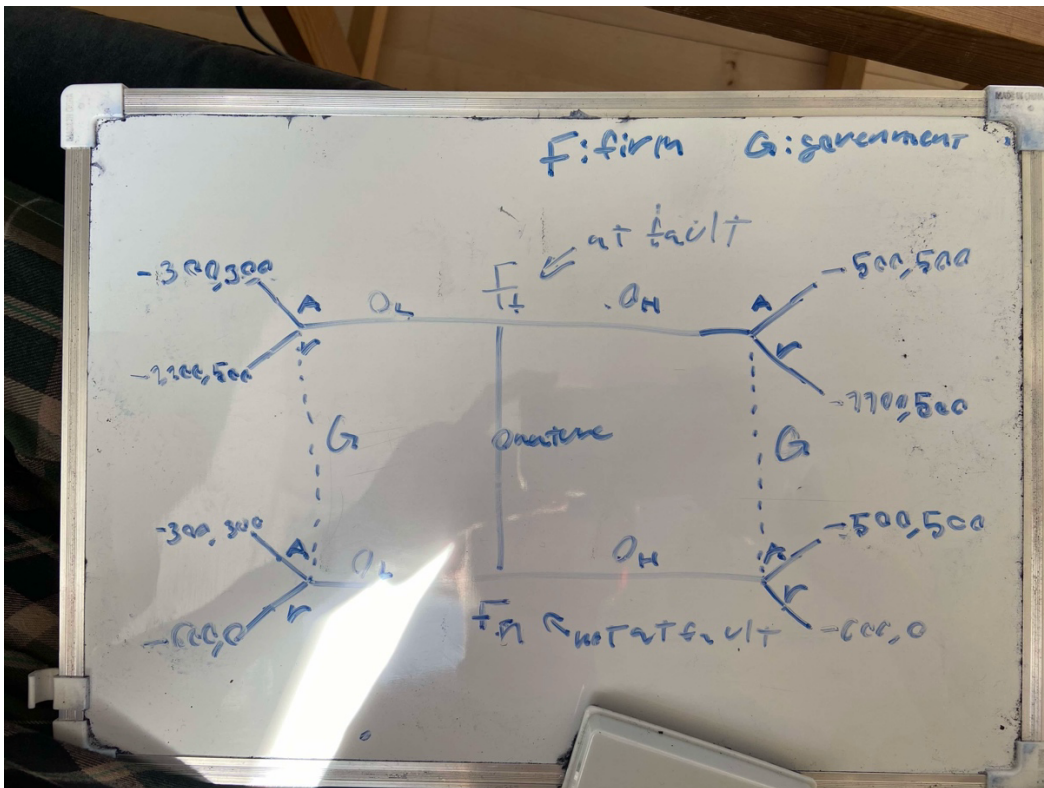


## Assignment 7

a)



b)

we check each starting with split EQ:

$(H, L)$

We notice that G accepts the low offer in the not guilty state, this means that the "at fault type will want to deviate to signal "low bid" and receive his lowest possible payoff being -300 thus this can never be an EQ

$(L, H)$

Given low bid G will reject

Given high bid G will accept

Giving us possible EQ =  $\{(L, H), (r, a)\}$

In this EQ F prefers to place a high bid given faulty giving him payoff -500 instead of -1100

Thus this cannot be an eq

Pooling:

$(H, H)$

$$\mu_H = 0.5$$

*G prefers A given H*

For F not to deviate G must not play A given low thus we find  $\mu_r$  such that

$$\mathbb{E}[u(A)|L] < \mathbb{E}[u(R)|L]$$

$$\mu_r \cdot 300 + (1 - \mu_r) \cdot 300 < 500 \cdot \mu_r$$

$$\Leftrightarrow \frac{3}{5} < \mu_r$$

Thus we have a pooling EQ

$$\sigma^* = ((Fault \rightarrow H, Not\ Fault \rightarrow H), (H \rightarrow A, L \rightarrow R), for\ \mu_H = \frac{1}{2}, \mu_R > \frac{3}{5})$$

$$\begin{aligned} & (L, L) \\ & \mu_R = \frac{1}{2} \\ & \mathbb{E}[u(A)|L] = 300 \\ & \mathbb{E}[u(R)|L] = \frac{1}{2} \cdot 500 = 250 \end{aligned}$$

Thus  $L \rightarrow A$

F gets -300 in payoff in both cases which are both the lowest possible meaning F would never deviate

c) check the intuitive criterion

$$(L, L)$$

In either state F gets his lowest possible payoff meaning that he would never deviate satisfying the intuitive criterion

$$((Fault \rightarrow H, Not\ Fault \rightarrow H), (H \rightarrow A, L \rightarrow R), for\ \mu_H = \frac{1}{2}, \mu_R > \frac{3}{5})$$

Both faulty and not faulty player types can obtain a larger payoff(given different moves by G) by deviating thus we can not set  $\mu_R$  to 0 or 1 as we would want to eliminate an EQ using the intuitive criterion

In konklusion:

The intuitive criterion removes no equilibria